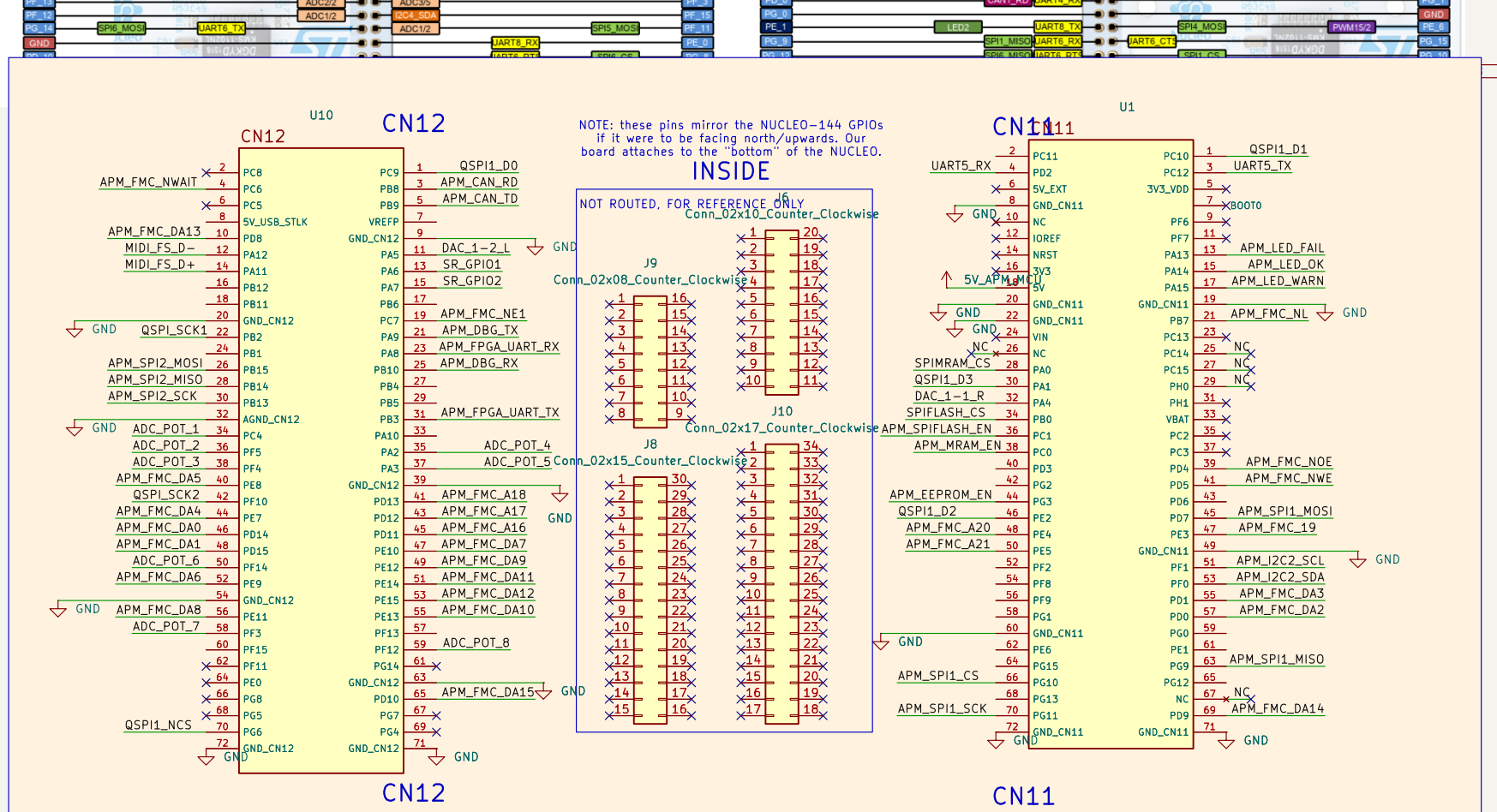
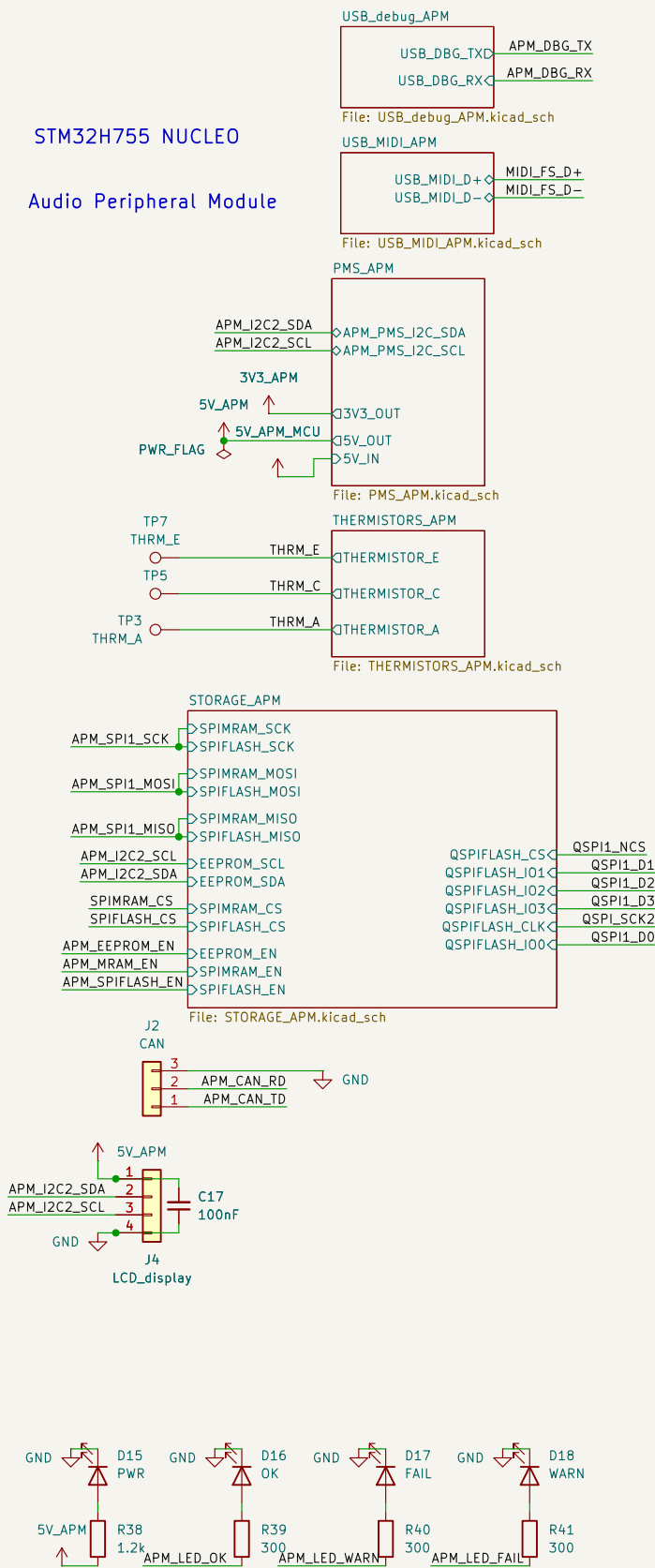


STM32H755 NUCLEO

Audio Peripheral Module

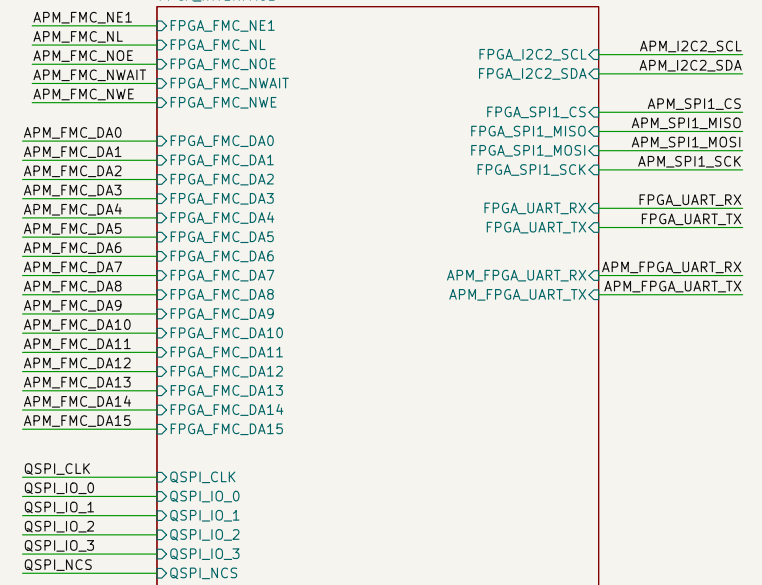


NOTE: these pins mirror the NUCLEO-144 GPIOs if it were to be facing north/upwards. Our board attaches to the "bottom" of the NUCLEO.

INSIDE

NOT ROUTED, FOR REFERENCE ONLY

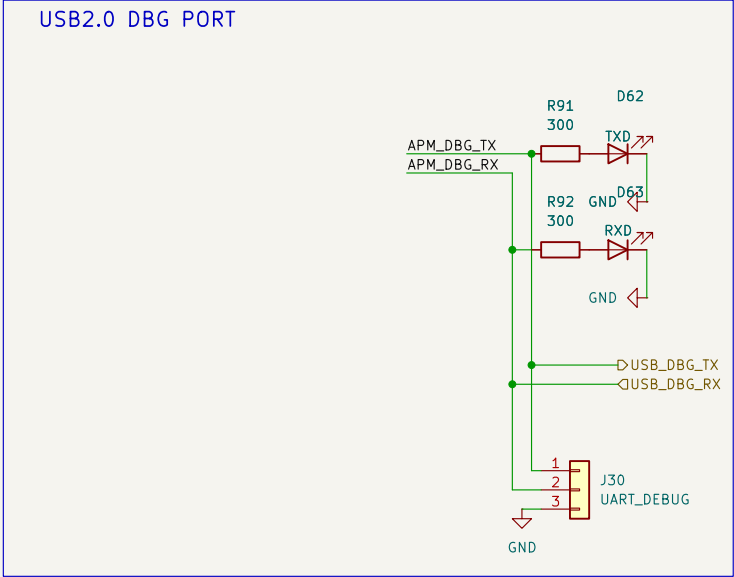
FPGA_INTERFACE



File: FPGA_INTERFACE.kicad_sch

UART -> USB Debug Port

UART debug messages available via a USB port and a 3 pin jumper for TX, RX, and GND UART connections



Sheet: /USB_debug_APM/
File: USB_debug_APM.kicad_sch

Title:

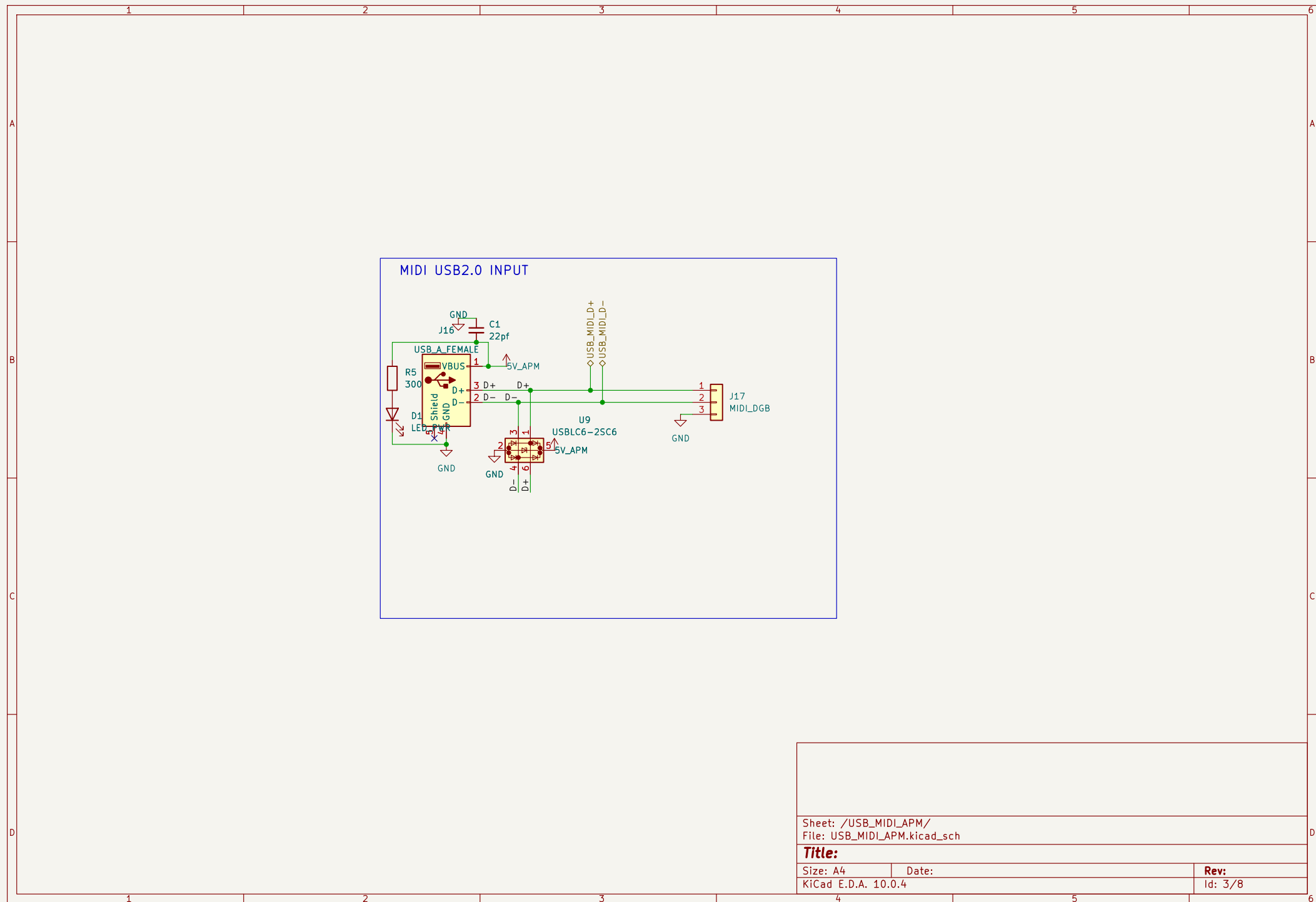
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Date:

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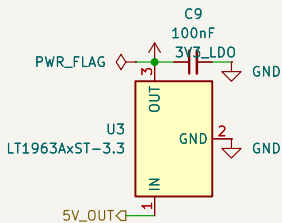
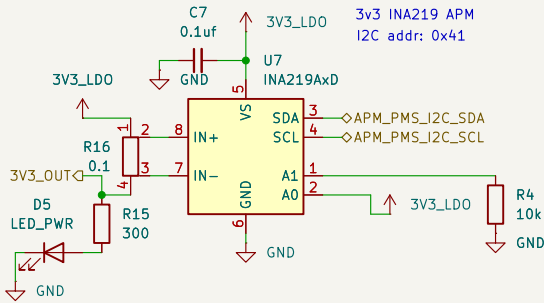
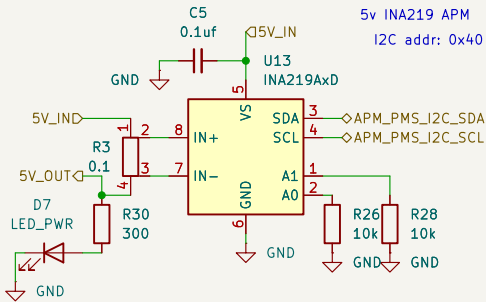
Rev:

Id: 2/8



Power Monitor and Supply

PMS has an INA219 current sensor
for each board/node in the
PCB. Each takes in the 5V from it's
parent board and supplies 5V to said
board's circuit (its peripherals)
thus enabling monitoring



Sheet: /PMS_APM/
File: PMS_APM.kicad_sch

Title:

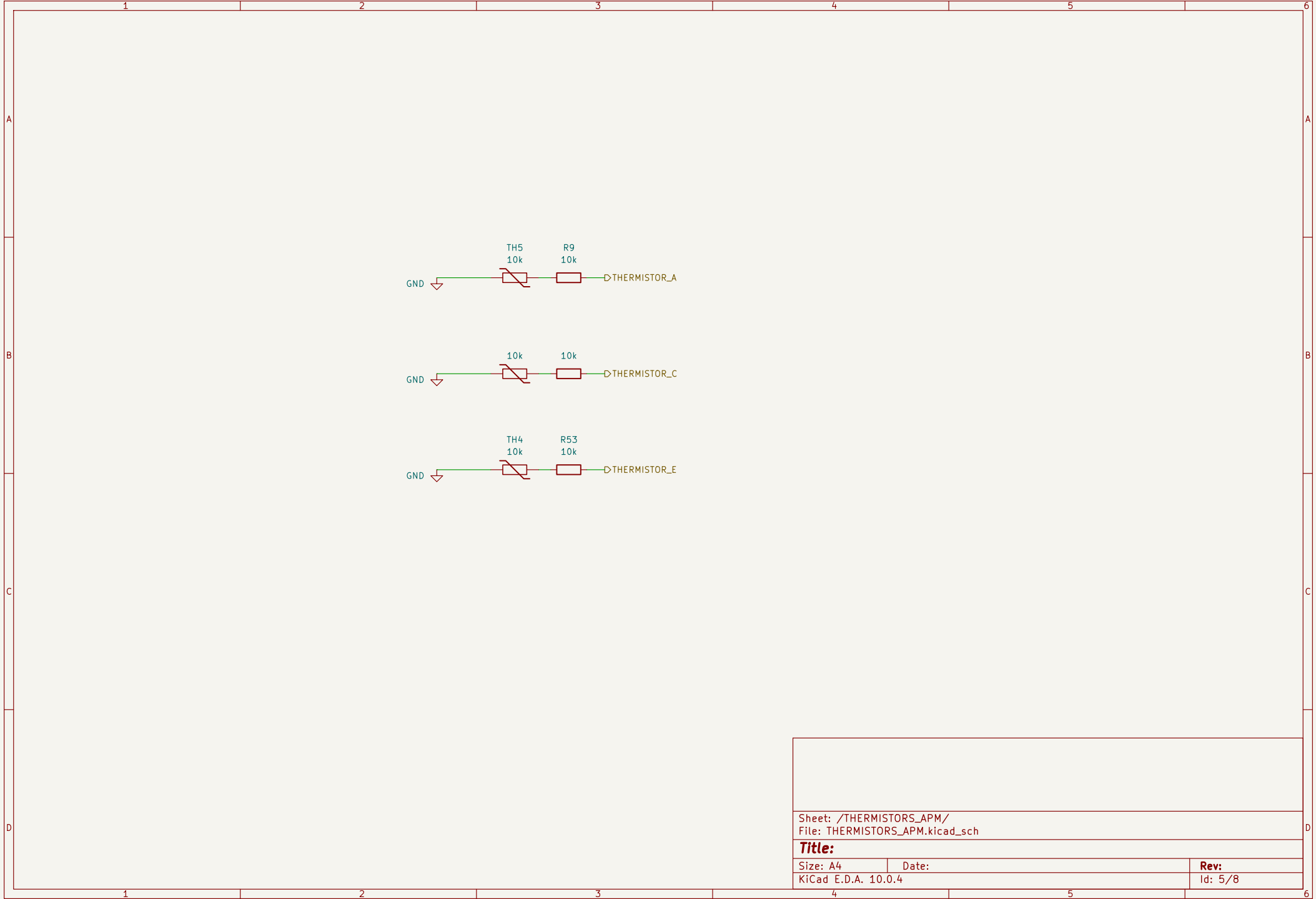
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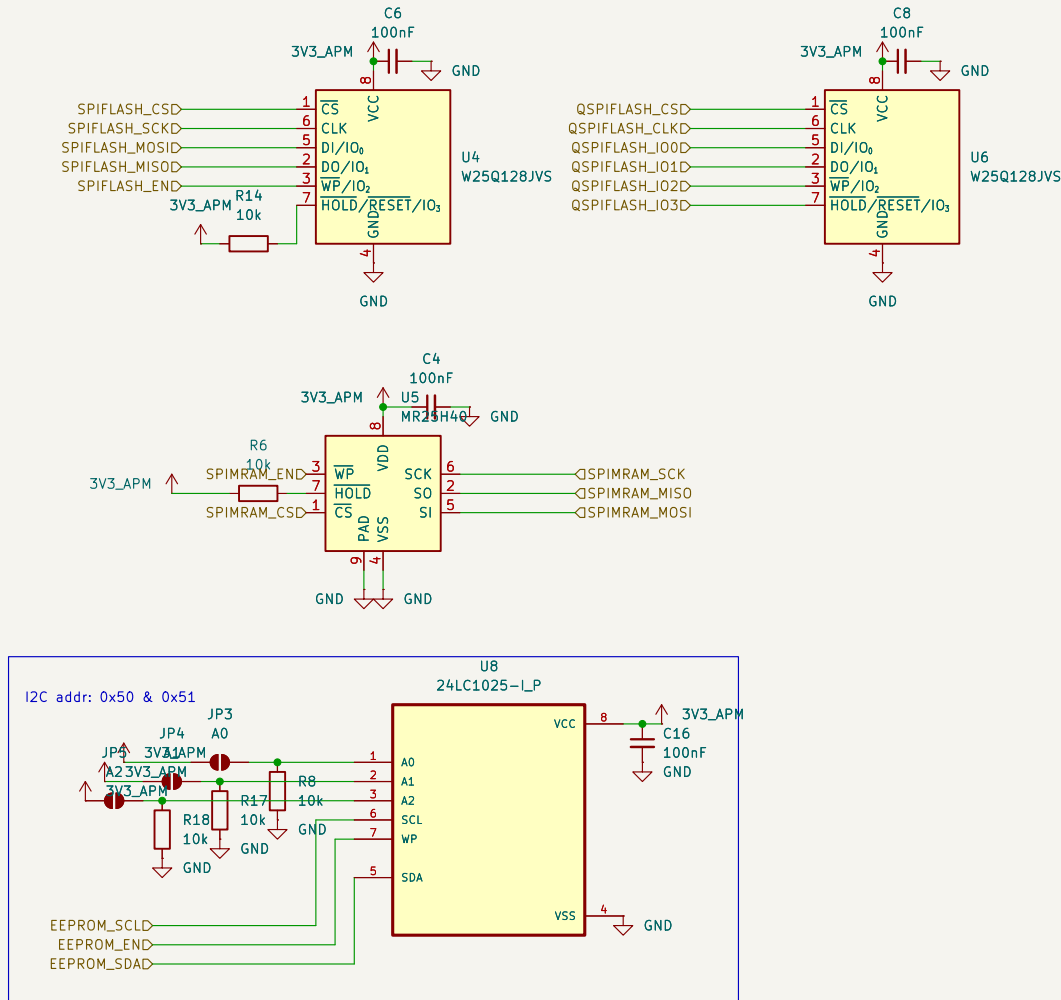
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Rev:

Id: 4/8





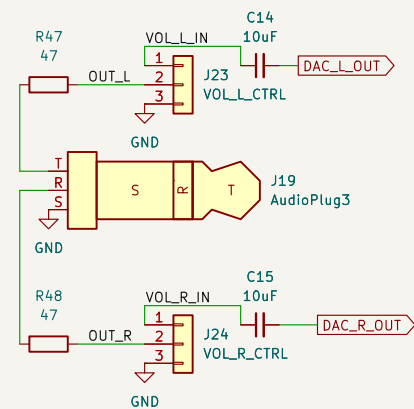
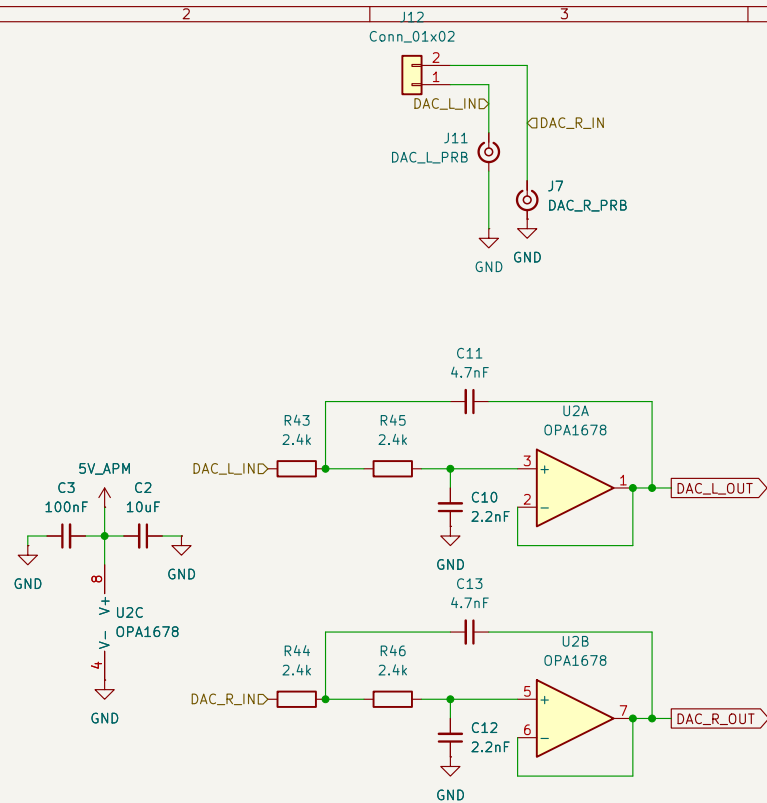
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File: STORAGE_APM.kicad_sch

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Size: A4 Date:

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Id: 6/8



Sheet: /AUDIO_APM/
File: AUDIO_APM.kicad_sch

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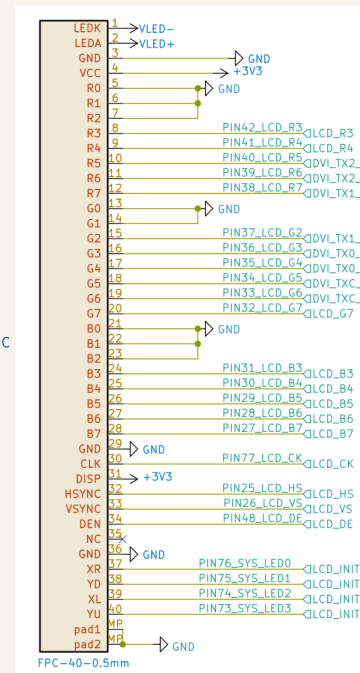
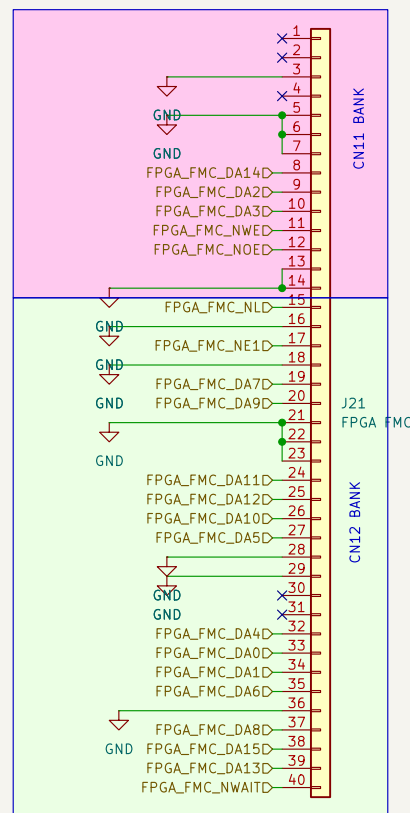
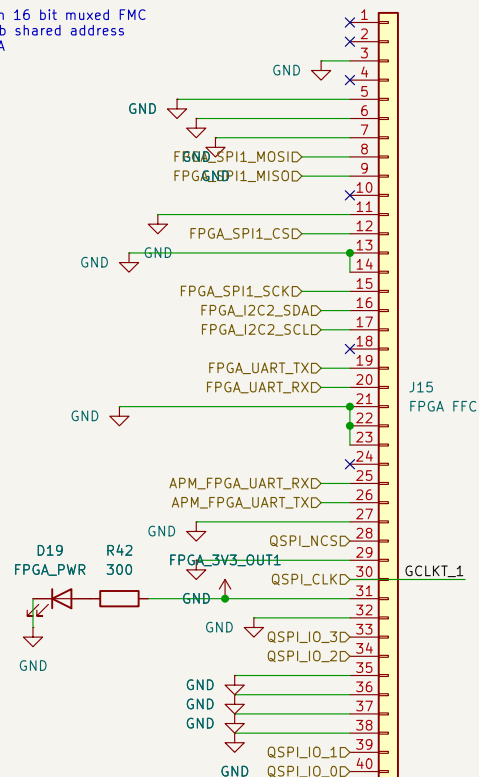
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Id: 7/8

Dual FPC STM32<-->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with 16 bit muxed FMC configuration resulting in a 128kb shared address space between the STM32 & FPGA



Sheet: /FPGA_INTERFACE/
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Id: 8/8